

scattering by a central potential

OCT 22 2024

INCIDENT ENERGY $E = \frac{m}{2} v_0^2$

INCIDENT VELOCITY $v_0 = \sqrt{2E/m}$

ANGULAR MOMENTUM $l = S m v_0 = S \sqrt{2mE}$

$$S = \frac{l}{\sqrt{2mE}}$$

APSIDAL
POINT



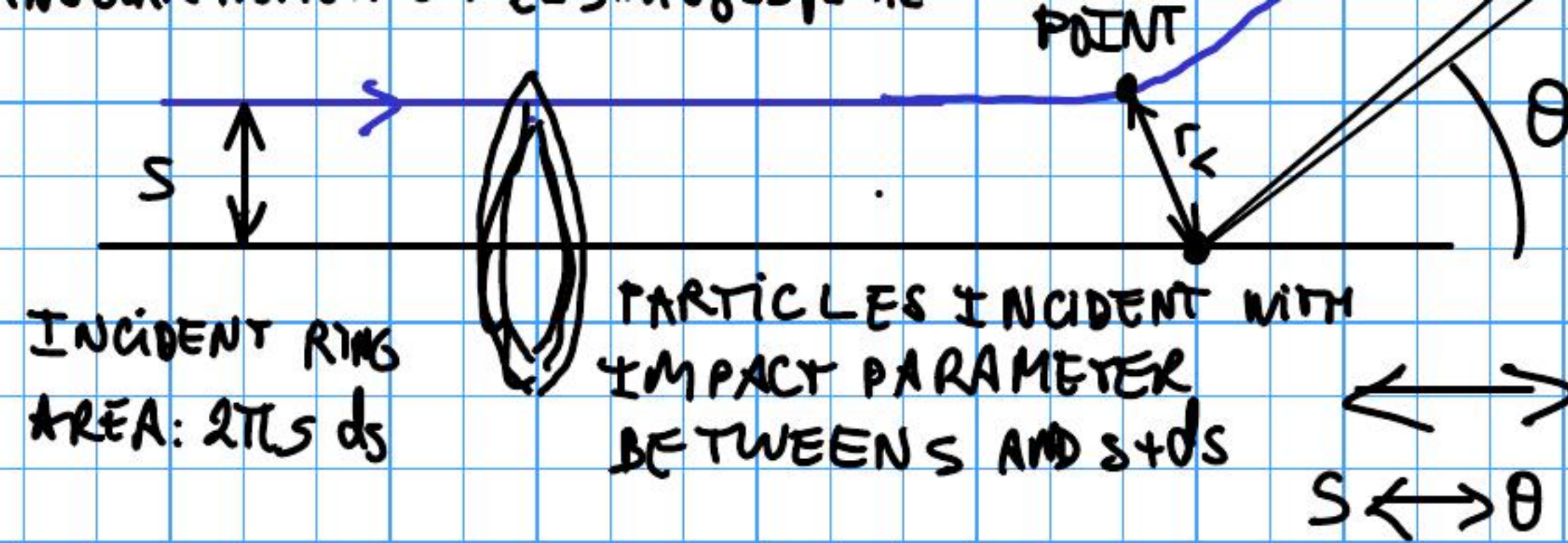
SCATTERED RING
SOLID ANGLE:

$$d^3r = r^2 dr \underbrace{d\phi \sin\theta d\theta}_{d\Omega}$$

$$d\Omega = 2\pi \sin\theta d\theta$$

Azimuthal angle
(ϕ irrelevant
by symmetry)

$$S(\theta, E)$$



INCIDENT RING
AREA: $2\pi s ds$

PARTICLES INCIDENT WITH
IMPACT PARAMETER
BETWEEN s AND $s+ds$

ϕ irrelevant
by symmetry

$S(\theta, E)$

PARTICLES EMERGING WITH SCATTERING
ANGLE BETWEEN θ AND $\theta+d\theta$

INCIDENT INTENSITY

$$I = \frac{\text{\# PART INCIDENT PER UNIT TIME}}{\text{TRANSVERSE BEAM CROSS SECTION AREA}}$$

DIFFERENTIAL CROSS SECTION

$$\sigma(\Omega) d\Omega = \frac{\text{\# PART SCATTERED INTO } d\Omega \text{ PER UNIT TIME}}{I}$$

INCIDENT INTENSITY

$$I = \frac{\text{\# PART INCIDENT PER UNIT TIME}}{\text{INCIDENT BEAM CROSS SECTION AREA}}$$

DIFFERENTIAL CROSS SECTION

$$\sigma(\Omega) d\Omega = \frac{\text{\# PART SCATTERED INTO } d\Omega \text{ PER UNIT TIME}}{\text{INCIDENT INTENSITY}}$$

$$I (2\pi s) |ds| = \text{\# part going through ring between } s \text{ and } s+ds = \text{\# part emerging with angle between } \theta \text{ and } \theta+d\theta = I \sigma(\theta) 2\pi \sin\theta |d\theta|$$

THEREFORE:

$$\sigma(\theta) = \frac{s}{\sin\theta} \left| \frac{ds}{d\theta} \right|$$

DEFINE ALSO THE TOTAL CROSS SECTION:

$$\sigma_T = \int d\Omega \sigma(\Omega)$$

THEREFORE:

$$\sigma(\theta) = \frac{s}{\sin\theta} \left| \frac{ds}{d\theta} \right|$$

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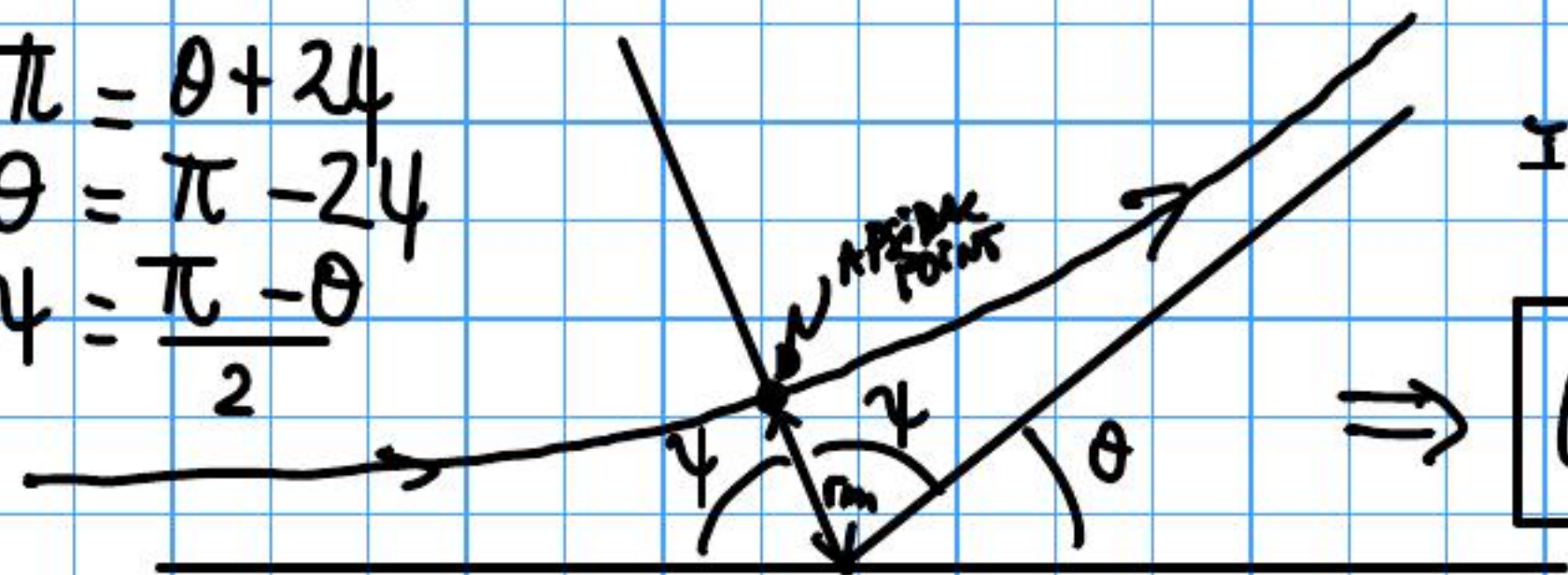
$$\sigma_T = \int d\Omega \sigma(\Omega)$$

REPULSIVE CASE:

$$\pi = \theta + 2\psi$$

$$\theta = \pi - 2\psi$$

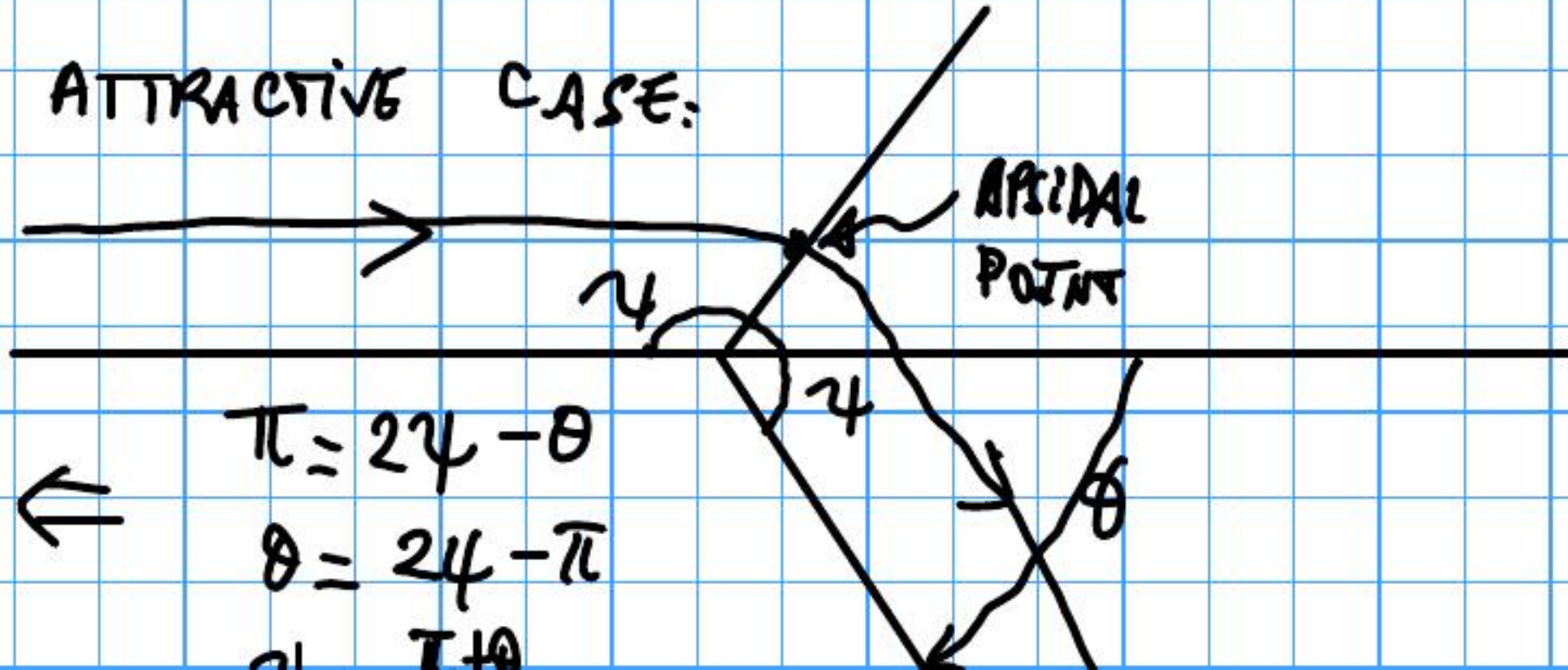
$$\psi = \frac{\pi - \theta}{2}$$



IN ALL CASES:

$$\theta = |\pi - 2\psi|$$

ATTRACTIVE CASE:



$$\pi = 2\psi - \theta$$

$$\theta = 2\psi - \pi$$

$$\psi = \frac{\pi + \theta}{2}$$

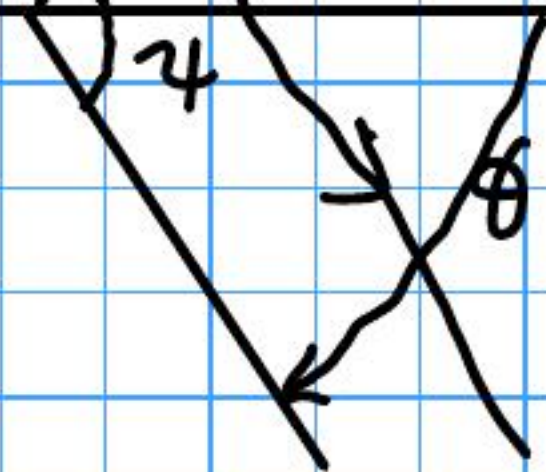
$$\psi = \frac{\pi - \theta}{2}$$


 $\Rightarrow$

$$\theta = |\pi - 2\psi|$$

 $\Leftarrow$

$$\begin{aligned}\pi &= 2\psi - \theta \\ \theta &= 2\psi - \pi \\ \psi &= \frac{\pi + \theta}{2}\end{aligned}$$



COMPUTING $S(\theta, E)$ (UNIVERSAL METHOD)

$$E = \frac{m}{2} \dot{r}^2 + V(r) + \frac{l^2}{2mr^2}$$

$$\dot{r} = \left[\frac{2}{m} \left(E - V(r) - \frac{l^2}{2mr^2} \right) \right]^{1/2}$$

$$\dot{r} = \frac{dr}{dt} \Rightarrow dt = \frac{dr}{\dot{r}}$$

$$l = mr^2 \dot{\theta} \Rightarrow \frac{d\theta}{dt} = \dot{\theta} = \frac{l}{mr^2} \Rightarrow d\theta = \dot{\theta} dt = \frac{l}{mr^2} dt = \frac{l}{mr^2} \frac{dr}{\dot{r}} = \frac{l}{mr^2} \frac{dr}{\left[\frac{2}{m} \left(E - V(r) - \frac{l^2}{2mr^2} \right) \right]^{1/2}} = \frac{l}{m} \sqrt{\frac{m}{2E}} \frac{r^{-2} dr}{\left[1 - \frac{V(r)}{E} - \frac{l^2}{2mE} \frac{1}{r^2} \right]^{1/2}} = \frac{l}{\sqrt{2mE}} \frac{dr}{r^2 \left[1 - \frac{V(r)}{E} - \frac{l^2}{2mE} \frac{1}{r^2} \right]^{1/2}}$$

$$\psi = \int_{\psi}^{\psi} d\psi = \int_{\pi}^{\pi} d\theta = \int_{\infty}^{\infty} S r^{-2} dr$$

$$\chi = \int_0^{\pi} d\chi = \int_{\theta_m}^{\pi} d\theta = \left[\int_{r_m}^{\infty} \frac{s r^{-2} dr}{\left[1 - V(r) - \frac{s^2}{r^2}\right]^{1/2}} \right] \Rightarrow \boxed{\theta(s) = \left| \pi - 2 \int_0^{u_m} \frac{s du}{\left[1 - \frac{V(u)}{E} - s^2 u^2\right]^{1/2}} \right|}$$

$u = 1/r \quad du = -r^{-2} dr \quad u_m = 1/r_m$

$\frac{1}{2m} \left(E - V(r) - \frac{L^2}{2mr^2} \right)^{1/2} = \frac{1}{\sqrt{2mE}} s \left[1 - \frac{V(r)}{E} - \frac{L^2}{2mE r^2} \right]^{1/2} = \frac{1}{\sqrt{2mE}} s$

From here, compute $\left| \frac{ds}{d\theta} \right| = \frac{1}{\left| \frac{d\theta}{ds} \right|}$ and use it to get $\sigma(\theta) = \frac{s}{\sin\theta} \left| \frac{ds}{d\theta} \right|$

UP TO HERE, RESULTS ARE VALID FOR ANY CENTRAL POTENTIAL $V(r)$

Coulomb Scattering

$$E > 0$$

hyperbolic

$$E = 0$$

parabolic

$$E < 0$$

elliptic

ONLY FOR $V = -\frac{k}{r}$!!!

FORCE

$$F = \frac{Z Z' e^2}{r^2}$$



$$f = -\frac{k}{r^2} \quad V = -\frac{k}{r}$$

$$k = -Z Z' e^2$$

Scattering: $E > 0$

$$\frac{1}{r} = G \left[1 + E \cos(\tilde{\theta} - \tilde{\theta}') \right]$$

$$G = \frac{m k}{l^2} = -\frac{m}{l^2} Z Z' e^2$$

$$\text{Eccentricity } E = \left[1 + \frac{2 E l^2}{m k^2} \right]^{\frac{1}{2}} = \left[1 + \frac{2 E l^2}{m (Z Z' e^2)^2} \right]^{\frac{1}{2}} = \left[1 + \left(\frac{2 s E}{Z Z' e^2} \right)^2 \right]^{\frac{1}{2}} > 1$$

$$l = s m v_0 = s \sqrt{2 m E} \rightarrow \frac{2 l^2 E}{m} = \frac{s^2 2^2 m E^2}{m} = (2 s E)^2$$

2 SITUATIONS:

i) REPULSIVE CASE

$$Z Z' > 0 \Rightarrow$$

$$k = -Z Z' e^2 < 0$$

2 SITUATIONS: i) REPULSIVE CASE $zz' > 0 \Rightarrow k = -zz'e^2 < 0$

ii) ATTRACTIVE CASE $zz' < 0 \Rightarrow k = -zz'e^2 > 0$ (ALSO GRAVITY $k = Gm_1m_2$)

i) REPULSIVE CASE: $k < 0$ $Q = \frac{mk}{l^2} < 0$

$$\frac{1}{r} = Q[1 + \epsilon \cos(\tilde{\theta} - \tilde{\theta}')] > 0 \Leftrightarrow 1 + \epsilon \cos(\tilde{\theta} - \tilde{\theta}') < 0 \Leftrightarrow \epsilon \cos(\tilde{\theta} - \tilde{\theta}') < -1 \Leftrightarrow \cos(\tilde{\theta} - \tilde{\theta}') < -\frac{1}{\epsilon}$$

We need the cosine to be negative and larger in magnitude than $1/\epsilon$

In this case, $\tilde{\theta} = \tilde{\theta}'$ is not a physical point, because $\cos(\tilde{\theta}' - \tilde{\theta}') = \cos(0) = 1 \Rightarrow \frac{1}{r} = Q[1 + \epsilon] < 0$

SO $\tilde{\theta} = \tilde{\theta}'$ IS IN THE MIDDLE OF THE INACCESSIBLE REGION OF THE HYPERBOLA, AND $\tilde{\theta} = \tilde{\theta}' \pm \pi$ WILL BE THE PERIAPSIS, SO WE CHOOSE $\tilde{\theta}' = \pi$ AND WE GET

UNPHYSICAL

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$$\cos(\tilde{\theta} - \tilde{\theta}') = \cos(\tilde{\theta} - \pi) = \cos \tilde{\theta} \underbrace{\cos \pi}_{-1} + \sin \tilde{\theta} \underbrace{\sin \pi}_0 = -\cos \tilde{\theta}$$

$$\frac{1}{r} = \left(-\frac{mk}{l^2}\right) [-1 - \epsilon \cos(\tilde{\theta} - \tilde{\theta}')] = \frac{mz z' e^2}{l^2} [\epsilon \cos \tilde{\theta} - 1]$$

MAXIMUM AT $\tilde{\theta} = 0 \Rightarrow$

MINIMUM r : $\frac{1}{r_m} = \frac{mz z' e^2}{l^2} [\epsilon - 1]$

SO THE ANGLE AT THE APSIDAL POINT IS $\tilde{\theta} = 0$

WE NOW NEED $\psi = \tilde{\theta}(r \rightarrow \infty) = \tilde{\theta}\left(\frac{1}{r} \rightarrow 0\right)$

[IN THE REPULSIVE CASE]
 $\theta = \pi - 2\psi \Rightarrow \psi = \frac{\pi - \theta}{2}$

$$0 = \epsilon \cos \psi - 1 \Leftrightarrow \frac{1}{\epsilon} = \cos \psi = \cos\left(\frac{\pi - \theta}{2}\right) = \cos \frac{\pi}{2} \cos \frac{\theta}{2} + \underbrace{\sin \frac{\pi}{2}}_1 \sin \frac{\theta}{2} = \sin \frac{\theta}{2}$$

WE NOW NEED $\psi = \tilde{\theta}(r \rightarrow \infty) = \tilde{\theta}\left(\frac{1}{r} \rightarrow 0\right)$ $\left[\theta = \pi - 2\psi \Rightarrow \psi = \frac{\pi - \theta}{2} \right]$

$$0 = \epsilon \cos \psi - 1 \Leftrightarrow \frac{1}{\epsilon} = \cos \psi = \cos\left(\frac{\pi - \theta}{2}\right) = \underbrace{\cos \frac{\pi}{2}}_0 \cos \frac{\theta}{2} + \underbrace{\sin \frac{\pi}{2}}_1 \sin \frac{\theta}{2} = \sin \frac{\theta}{2}$$

ii) ATTRACTIVE CASE: $k > 0$ $C = \frac{km}{l^2} > 0$

TAKE $\tilde{\theta}' = 0$ $\tilde{\theta} = \tilde{\theta}' = 0 \Rightarrow \frac{1}{r} = C[1 + \epsilon \cos(0)] = C[1 + \epsilon]$

APSIDAL POINT IS ALSO AT $\tilde{\theta} = 0$

WE NOW NEED $\psi = \tilde{\theta}(r \rightarrow \infty) = \tilde{\theta}\left(\frac{1}{r} \rightarrow 0\right)$

$$0 = 1 + \epsilon \cos \psi \Leftrightarrow -\frac{1}{\epsilon} = \cos \psi = \cos\left(\frac{\pi + \theta}{2}\right) = \underbrace{\cos \frac{\pi}{2}}_0 \cos \frac{\theta}{2} - \underbrace{\sin \frac{\pi}{2}}_1 \sin \frac{\theta}{2} = -\sin \frac{\theta}{2}$$

MAX AT $\tilde{\theta} = 0$

MINIMUM r : $\frac{1}{r_m} = C[1 + \epsilon]$

$\left[\text{IN THE ATTRACTIVE CASE} \right]$
 $\left[\theta = 2\psi - \pi \Rightarrow \psi = \frac{\pi + \theta}{2} \right]$

We now combine both cases:

$$\cot^2 \frac{\theta}{2} = \frac{1}{\tan^2 \frac{\theta}{2}} = \frac{\cos^2 \frac{\theta}{2}}{\sin^2 \frac{\theta}{2}} = \frac{1 - \sin^2 \frac{\theta}{2}}{\sin^2 \frac{\theta}{2}} = \frac{1}{\sin^2 \frac{\theta}{2}} - 1 = \epsilon^2 - 1 = \left(\frac{2sE}{2Z' e^2} \right)^2 \Rightarrow \boxed{s = s(\theta, E) = \left| \frac{2Z' e^2}{2E} \right| \cot \frac{\theta}{2}}$$

$$\frac{ds}{d\theta} = \left| \frac{2Z' e^2}{2E} \right| \frac{d}{d\theta} \left(\frac{\cos \frac{\theta}{2}}{\sin \frac{\theta}{2}} \right) = \left| \frac{2Z' e^2}{2E} \right| \left(\frac{1}{2} \left(\frac{-\sin \frac{\theta}{2}}{\sin^2 \frac{\theta}{2}} \right) - \frac{1}{2} \frac{\cos \frac{\theta}{2}}{\sin^2 \frac{\theta}{2}} \right) = - \left| \frac{2Z' e^2}{4E} \right| \frac{1}{\sin^2 \frac{\theta}{2}}$$

$$\sigma(\theta) = \frac{s}{\sin \theta} \left| \frac{ds}{d\theta} \right| = \frac{1}{2} \left(\frac{2Z' e^2}{2E} \right)^2 \frac{1}{2 \sin \frac{\theta}{2} \cos \frac{\theta}{2}} \frac{\cancel{\cos \frac{\theta}{2}}}{\sin \frac{\theta}{2}} \frac{1}{\sin^2 \frac{\theta}{2}}$$

$$\frac{ds}{d\theta} = \left| \frac{Z Z' e^2}{2E} \right| \frac{d}{d\theta} \left(\frac{\cos \frac{\theta}{2}}{\sin^2 \frac{\theta}{2}} \right) = \left| \frac{Z Z' e^2}{2E} \right| \left(\frac{1}{2} \left(\frac{-\sin \frac{\theta}{2}}{\sin^3 \frac{\theta}{2}} \right) - \frac{1}{2} \frac{\cos \frac{\theta}{2}}{\sin^2 \frac{\theta}{2}} \right) = - \left| \frac{Z Z' e^2}{4E} \right| \frac{1}{\sin^2 \frac{\theta}{2}}$$

$$\sigma(\theta) = \frac{s}{\sin \theta} \left| \frac{ds}{d\theta} \right| = \frac{1}{2} \left(\frac{Z Z' e^2}{2E} \right)^2 \frac{1}{\sin^2 \frac{\theta}{2} \cos \frac{\theta}{2} \sin \frac{\theta}{2}} \frac{1}{\sin^2 \frac{\theta}{2}}$$

$$\sigma(\theta) = \frac{1}{4} \left(\frac{Z Z' e^2}{2E} \right)^2 \csc^4 \left(\frac{\theta}{2} \right)$$

Rutherford scattering cross section
(for Coulomb forces)

Remarks:

1) The cross section is the same for the repulsive and attractive cases, since it is proportional to $(Z Z')^2$.

Remarks:

- 1) The cross section is the same for the repulsive and attractive cases, since it is proportional to $(zz')^2$.
- 2) It diverges like E^{-2} at low energies.
- 3) It diverges like θ^{-4} (VERY STRONG!!!) for small angles θ .
- 4) This classical result is exactly the same as the result obtained in Quantum Mechanics.

Total scattering cross section:

NOTATION:

σ_{tot}

σ_{el}

Total scattering cross section:

$$\sigma_T = \int \sigma(\Omega) d\Omega$$

In this case $\sigma(\theta) = \text{const} \frac{1}{\sin^4(\frac{\theta}{2})}$

$$d\Omega = 2\pi \sin\theta d\theta = 4\pi \sin\frac{\theta}{2} \cos\frac{\theta}{2} d\theta$$

NOTATION:

Goldstein

Others

$$\sigma(\Omega) \leftrightarrow \frac{d\sigma}{d\Omega}$$

DIFFERENTIAL SCATTERING CROSS SECTION

$$\sigma_T \leftrightarrow \sigma \text{ TOTAL SCATTERING CROSS SECTION}$$

STRONGLY

$$d\Omega = 2\pi \sin\theta d\theta = 4\pi \sin\frac{\theta}{2} \cos\frac{\theta}{2} d\theta$$

$$\sigma_T = \frac{4\pi}{4} \left(\frac{Z Z' e^2}{2E} \right)^2 \int_0^\pi \frac{\sin\frac{\theta}{2} \cos\frac{\theta}{2}}{\sin^4\frac{\theta}{2}} d\theta$$

$$\sim \int_0^\pi \frac{d\theta}{(\theta/2)^3}$$

$$= 8 \int_0^\pi \frac{d\theta}{\theta^3} = 8 \left. \frac{\theta^{-2}}{(-2)} \right|_0^\pi = +\infty \quad \text{STRONGLY DIVERGENT}$$

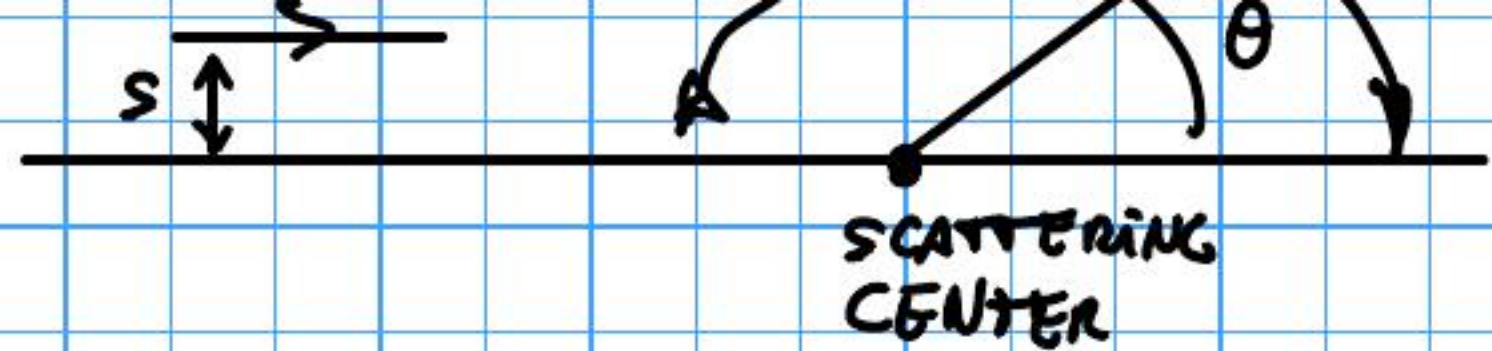


CLASSICAL

QUANTUM

$\sigma_T = +\infty$ for any long range potential

$\sigma_T = +\infty$ for some long range potentials
 $V(r)$ do not decay fast enough at $r \rightarrow \infty$



QUANTUM

$a_T = +\infty$ for some long range potentials
 $V(r)$ do not decay fast enough at $r \rightarrow \infty$

CAVEAT

1) IN A DENSE
 CONDUCTING MATERIAL:

$$V = -\frac{k}{r} \longrightarrow V_{\text{SCREEN}}(r) = -\frac{k}{r} e^{-r/\lambda}$$

SCREENED COULOMB
 POTENTIAL ("YUKAWA")

(In non-conducting materials $V_{\text{SCREEN}}(r)$ may take a different form, but still will vanish much faster than $1/r$)

Chapter 8 Hamiltonian and Hamilton's Eqs of Motion

$$L = T - V = L(q_1, \dots, q_n, \dot{q}_1, \dots, \dot{q}_n, t)$$

$$0 = \frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}_i} \right) - \frac{\partial L}{\partial q_i} \quad p_i = \frac{\partial L}{\partial \dot{q}_i}$$

MIDTERM REMARK

In polar coordinates

$$\frac{\partial L}{\partial \dot{\theta}} = p_\theta = l_z$$

~~$$L = L(r, \dot{r}, p_\theta)$$~~

Lawrence Evans

$$\frac{\partial q_i}{\partial q_j} \bigg|_{q_j}$$

$$\frac{\partial q_i}{\partial q_j} \bigg|_{q_j}$$

$$\frac{\partial q_i}{\partial q_j} \bigg|_{q_j}$$

~~Lagrange Eqs.~~

Legendre Transformation $q_1, \dots, q_n, \dot{q}_1, \dots, \dot{q}_n \rightarrow q_1, \dots, q_n, p_1, \dots, p_n$

$$f(x, y) \Rightarrow df = \frac{\partial f}{\partial x} dx + \frac{\partial f}{\partial y} dy$$

In Thermodynamics

Fluid at constant N

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Fluid at constant N

1st Law $dU = dQ + dW = T dS - p dV$

$$U = U(S, V)$$

Internal Energy

We want T, V as variables

$F = A = U - TS$ Helmholtz Free Energy

$$dA = dU - d(TS) = \cancel{T dS} - p dV - \cancel{T dS} - S dT = -S dT - p dV \Rightarrow A = A(T, V)$$

$$dA = dU - d(TS) = \cancel{TdS} - p dV - \cancel{TdS} - S dT = -S dT - p dV \Rightarrow A = A(T, V)$$

In general

$$f(x, y) = u dx + v dy \quad u = \frac{\partial f}{\partial x} \quad v = \frac{\partial f}{\partial y} \quad (x, y) \longrightarrow (u, y)$$

$$g = f - ux \Rightarrow dg = df - d(ux) = \cancel{u dx} + v dy - \cancel{u dx} - x du = -x du + v dy$$

$$g = g(u, y)$$

$$g = g(u, y)$$

let's look at $L(q_1, \dots, q_n, \dot{q}_1, \dots, \dot{q}_n)$

$$p_i = \left. \frac{\partial L}{\partial \dot{q}_i} \right|_{\substack{q_1 \\ \vdots \\ q_n}}$$

$$H = - \left(L - \sum_i p_i \dot{q}_i \right)$$

$$dH = -dL + \sum_i d(p_i \dot{q}_i) = - \sum_i \left(\frac{\partial L}{\partial q_0} dq_0 + \frac{\partial L}{\partial \dot{q}_i} d\dot{q}_i \right) + \sum_i \left(p_i d\dot{q}_i + \dot{q}_i dp_i \right)$$

$-\frac{\partial L}{\partial t} +$

$$H = -\left(L - \sum_i p_i \dot{q}_i\right) \quad dH = -dL + \sum_i d(p_i \dot{q}_i) = -\sum_i \left(\frac{\partial L}{\partial q_i} dq_i + \cancel{\frac{\partial L}{\partial \dot{q}_i} d\dot{q}_i}\right) + \sum_i \left(\cancel{p_i d\dot{q}_i} + \dot{q}_i dp_i\right)$$

$$dH = -\frac{\partial L}{\partial t} dt + \sum_i \left(\dot{q}_i dp_i - \frac{\partial L}{\partial q_i} dq_i\right) \Rightarrow H = H(p_1, \dots, p_n, q_1, \dots, q_n, t)$$

$$\boxed{\begin{aligned} \frac{\partial H}{\partial p_i} &= \dot{q}_i & \frac{\partial H}{\partial q_i} &= -\frac{\partial L}{\partial q_i} \stackrel{\text{Lagrange Eqn}}{=} -\dot{p}_i & \frac{\partial H}{\partial t} &= -\frac{\partial L}{\partial t} \end{aligned}}$$

CANONICAL EQS OF MOTION